Problem Statement

Identity Drift Detection in Autonomous AI Agents

Organizations now deploy AI agents that work on their own: browsing, reading data, calling APIs, and acting on behalf of employees, often without human approval at every step. Each agent is given its own identity, with a token and a set of permissions, so it can do its job.

This exposes a critical, unsolved gap: **security systems check whether an agent's credentials are valid, but not whether the agent is still behaving like the role it was created for.** A hidden instruction in a document, a poisoned tool, or a stolen token can change what an agent does while its identity stays fully valid, so every access check still passes.

Build a system that learns the normal behavior of AI agents, continuously compares it with each agent's declared role, and detects when an agent's behavior no longer matches that role, even though its credentials are still valid. The system should explain every alert with a clear risk score, and restrict or isolate the agent without disrupting legitimate work.

Background

Companies are giving AI agents their own accounts, API keys, and permissions, much as they do for employees. Traditional identity and access management was built for humans and fixed-purpose software, and it asks one question: *"Is this identity allowed to do this?"*

AI agents are different. They decide what to do at runtime, they read untrusted content such as emails, web pages, and documents, and they are often given broader access than their job needs.

Example: A Procurement Agent normally checks supplier prices and creates purchase requests. After reading a supplier's PDF with a hidden instruction, it starts reading HR salary files and sending data to an unknown external server. Its token and permissions are still valid, so normal security tools raise no alarm.

Standards bodies and security vendors are now working on agent identity, authorization, and behavior monitoring, and recent industry surveys report that many organizations have had security incidents involving AI agents. Detecting this kind of change in behavior reliably is still an open problem.

Core Challenges

- **Valid credentials hide the attack:** the agent is authorized, so permission checks alone can't tell a hijacked agent from a healthy one.

- **Indirect prompt injection:** malicious instructions buried in documents, web content, or messages from other agents look like normal input at face value.
- **Legitimate change vs. adversarial drift:** agents get new tasks, tools, and updates over time, so the system must tell *natural change* from *compromise* without over-triggering false alarms.
- **Little or no history:** many agents are short-lived or newly created, so there may be no past behavior to learn from.
- **Chains of agents:** one agent can call another, so responsibility and authority can become unclear, and a single compromised agent may influence others.
- **Evasion:** a smart attacker may copy normal behavior or change it slowly to stay under the detection threshold.